

ORIGINAL RESEARCH

A novel cyber-attack modelling and detection in overcurrent protection relays based on wavelet signature analysis

Mohammad Yousefi kia  | Mohsen Saniei  | Seyyed Ghodrattollah Seifossadat 

Department of Electrical Engineering, Faculty of Engineering, Shahid Chamran University of Ahvaz, Ahvaz, Iran

Correspondence

Mohsen Saniei, Department of Electrical Engineering, Faculty of Engineering, Shahid Chamran University of Ahvaz, Ahvaz 61357-43337, Iran.
Email: m.saniei@scu.ac.ir

Abstract

Cyber attacks can disrupt the digital protection relays and change their output command to the circuit breaker. The aim of this research is to provide new condition monitoring for overcurrent relays during cyber treatments. This research is modeled three common types of attacks with randomize concept and its detection method via MATLAB/SIMULINK software into IEEE 3-machine 9-bus model that is applicable in real time situation. Finally a new index has been defined for detection that employs wavelet analysis of the substation currents and its energy which has been called cyber attack detection index (CADI). Proposed index is robust to variation of sampling signal, sampling rate, cyber attack type and current transformer saturation. The results show that the index has an acceptable performance with accuracy of 100% without any false trigger. This method is not dependent on operating point, relay protection scheme and attack type and also could distinguish the abnormal actions of breakers due to unexpected cyber contingencies caused by natural disasters or software malfunctions rather than cyber attacks in a certain operating points. The level of CADI curve in cyber attacks is always higher than symmetrical fault and current signal detects them more quickly and clearly than other signals.

1 | INTRODUCTION

In recent years, with the smartness of modern power networks, they have become more complicated than before, so that the independent system operator faces more challenges in order to maintain the stability and reliability of the system in all three levels of generation, transmission and distribution. At the same time developing telecommunication systems cause to improve the performance of condition monitoring, protection and better control of the physical system and in this regard, they have added new advanced measurement infrastructure such as PMU¹ and PDC² to the grid [1]. Because cyber systems in cyber physical network guarantor of the efficiency and safe operation of the power network, but by looking at the history of blackouts in the power network, it can be seen that most of these

blackouts are due to the failures and breakdowns of this system, and the most important weakness and vulnerability of those are cyber-attacks [2]. Therefore, the issue of cyber security of smart grid is one of the most important challenges facing it, and the lack of attention to this issue can cause economic losses, damage to equipment, and finally extensive and cascading blackouts throughout the power system [3]. The main purpose of cyber-attacks includes two physical parts such as intelligent electronic devices and the cyber part including telecommunication, computing and software layers [4]. In general, cyber-attacks are destructive events that cause inappropriate behaviour, including improper functioning of the human-machine interface, control systems, tripping of machines, load shedding and cascading failure of the system [5]. Some basic components are services in modern power grid which include PLC³, SCADA⁴, DCS⁵ and

¹ Phasor measurement unit

² Phasor data concentrator

³ Programmable logic controllers

⁴ Supervisory control and data acquisition

⁵ Distributed control systems

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devices such as smart meters, RTU⁶ and intelligent field instruments including remotely programmable valves, intelligent electronic devices (relays and circuit breakers) are now connected via network and communication platform [6]. Various techniques and challenges related to the cyber security of power distribution networks, including their complexity and scale in the future, new telecommunication requirements, threats and vulnerable points of common communication systems, delay factors, heterogeneous protocols and brands, old devices, secure two-way communication and user privacy is stated in ref. [7].

This paper first investigates and addresses the false data injection (FDI) attacks and denial of service (DOS) on the power system, which target the protection systems. Attacks were modelled based on their concept in MATLAB/SIMULINK software. By launching the attacks and having access to system measurements, current signals are captured at related location in the power system and wavelet analysis is used to extract their features. Finally a robust detection technique is proposed based on energy of the wavelet details signals against the cyber-physical threats to detect the cyber-attacks.

2 | CYBER-SECURITY CONCEPT ON POWER SYSTEM

In general, factors such as availability, confidentiality, integrity, authentication and authorization are needed to reduce and deal with power system threats. In order to achieve an impenetrable power system, it is necessary to satisfy the three following security parameters famously known as CIA (confidentiality, integrity, and availability) triad [8] as described below:

- **Confidentiality:** Is the property, that information is not made available or disclosed to unauthorized individuals, entities or processes [9]. The confidentiality of data is manipulated by two kinds of cyber-attacks, including man in the middle and rouge device which affect measuring signal and command in remote or physical access. In a specific example it includes eavesdropping attacks, which interpret subscribers' data and analyse their consumption patterns. Therefore, this parameter is used in two fields of privacy and security.
- **Integrity:** Is the property of maintaining and assuring the accuracy and completeness of data over its entire life-cycle or protecting information from being modified by unauthorized parties [10].
- **Availability:** Is the property to ensure that authorized parties or entities able to access the information or devices when needed [11].

If any of these security parameters are not satisfied, the system is under a threat or attacked. In order to protect the smart power system against cyber threats, it is essential to analyse the

network architecture and its communications. This article investigates some basic points related to security threats in digital protective relay and circuit breaker. The most commonly attack models in literature include false data injection to the computer-aided system operating devices, DoS on the device, malicious switching behaviour of circuit breakers (CBs)/isolators. The major classification of cyber-attack in digital relay is described as below:

- **Man in the middle attacks:** By accessing the communication channels and changing the measurement devices, the attacker compromises two parameters of confidentiality and integrity. The implementation details and effects of these attacks are presented in [12]. A man-in-the-middle attack on GOOSE⁷ messages in the IEC 61850 communication protocol is demonstrated on protective relays installed in substations. Signature analysis of voltage and active power signals are used for cyber detection. The application of the RSA⁸ in this algorithm makes it unsuitable for applications that require time performance of 4 ms or less [13]. A special form of these attacks with catastrophic effects is known as a replay attack [14].
- **Rogue devices:** In this type of threats, the intruder plans the attack by having physical access to the equipment such as PMUs and replacing it with a rouge one, and in this way it affects the performance of the system by sending false signals and commands [15].
- **Denial of service (DoS):** The main purpose of service interruption attacks is to disrupt the access feature, which the attacker initiates by delaying or blocking communication links and sending a large amount of data with the aim of creating traffic [16]. There is a possibility of DOS attacks occurring in different communication layers of power systems, in the form of jamming in the channel for the physical layer with the effect of delay or failure to deliver the message, as well as changing MAC address parameters in the way of data spoofing [17]. A threat detection method on an IEC61850 based automated substation is proposed in [18]. Examining the weaknesses and vulnerabilities of various real hardware and software parts against DoS attacks and their effects on other layers, including transmission and network, is stated in [19]. The mentioned examples of DOS attacks include spoofing, flooding, and jamming, with the end result of delaying or failure to send time-dependent messages, causing system and equipment malfunctions and making impossible to communicate with devices. Investigating the severity of malicious switching and DoS attacks in protective digital relays for the Ukrainian power grid has been done by simulating cyber-attacks in aurora generator test [20].
- **False data injection (FDI):** This kind of attacks start by injecting false data with the aim of performing false control operations, which in the most well-known case are aimed at attacking the state estimation vector and bypassing the bad data detection methods, was first published in [21].

⁶ Remote telemetry units

⁷ Generic object oriented substation event

⁸ Rivest–Shamir–Adleman

In recent years, many studies have been focused on the scope of these attacks that the injection of incomplete information, cyber-attacks on the automatic generation control system and electricity market parameters are the most famous [22]. Several techniques have been developed to detect and reduce the effect of such attacks [23].

The effect of cyber-attacks on smart grid is very diverse in terms of the severity of the damage, and it ranges from small cases of energy theft to widespread blackouts and destruction of equipment, including engines and generators. Some examples of cyber-attacks with a high impact on the power system are stated in [24].

A cyber-attack is initiated by a hacker or an infiltrating organization, and the intruder can compromise the power system by penetrating through hardware, software or protocol vulnerabilities. The attacker takes advantage of the weakness of the physical and cyber security system by collecting information after a complete reconnaissance of the target and its components and executes the attack at the specific level of generation, transmission or distribution by deceiving control systems or protective devices. Protective relays, beside the functioning in fault conditions, have the possibility of being opened and closed remotely by the system operator with the help of communication channels, so if this wireless network is insecure, it facilitates the possibility of an invade [25]. In recent years, many studies have been conducted in the field of increasing cyber security in power networks and against cyber threats, which are mainly based on steady state analysis. The false data injection attack can lead to the distortion of the state estimation algorithm by having the topology information of a system before the attack [26].

An overview of the types of FDI attacks and their detection methods based on the leaked system topology information is described in [27]. An innovative method using local information to prevent undetectable attacks by weak intruder is presented [28]. A mathematical basis is formulated to quantify the economic effects of the occurrence of topological data attacks in an electricity market when the attacker uses a virtual bidding strategy [29]. After that, by analysing the steady state on the transmission line, the ability of the attackers to penetrate into the virtual bidding process for certain buses is calculated. In [30], a resilient method for identifying the FDI attack in line current differential relays is described, which is only specific to this type of relay. The author used two computational modules, positive sequence of PS and negative sequence of NS to calculate the voltage in terminal and compare with the measured voltage and validate the measured currents. Long computing procedures and the possibility of violating the permitted time limits for the operation of protective equipment is mentioned as its drawback. A meta-heuristic optimization algorithm is used in steady state to deal with three types of cyber-attacks [31]. Since steady state analysis is based on load flow, it cannot comprehensively express the non-linear behaviour and dynamics of the system. In the load flow calculation, the frequency is assumed to be constant, while the frequency constantly has dynamic changes. On the other hand, steady state analysis as an opera-

tional solution after a disturbance is unable to show the time sequence of switching events. In [32], FDI on the substation automation system include distance relays and power circuit breakers have been performed. A resilient protection scheme against cyber-attacks is proposed by utilization protection coordination principles to investigate protection setting changes and using correlation coefficient algorithm based on transient fault signature (TFS) in which the weaknesses in overcurrent and directional relays are eliminated. This reference implemented model regardless of practical issues related to substations. In [33], the effects of cyber-physical attacks in the power sector during subversive manipulation of communication paths and the performance of distributed generation resources are presented. In ref. [34], the dynamic effects of cyber-attacks on substations have been investigated and dynamic studies have been stated as an effective solution to detect cyber-attacks. This study only investigated the dynamic effects of switching attacks and did not deal with other attacks. In [35], with the help of analysing the relationship between relay settings and circuit breaker logic during the attack, a new method of risk assessment has been introduced. This evaluation method is only used when attacks occur and is not effective during conventional physical faults. It is noticeable that the behaviour of power networks is different in the two cases of cyber-attack and short circuit fault and requires different assumptions for risk assessment. In [36], FDI, switching, DoS and replay attacks have been applied to protection system (overcurrent relays), intelligent electronic device and power circuit breakers. The detection method is based on dynamic state analysis which is costly in real time. In [37], risk assessment of power systems in cyberattacks considering the role of protection system such as relay settings and parameters has been considered. ELC⁹ index is used for detection but the reason for choosing this index and its superiority over other methods dose not clearly mentioned. In [38], detection of cyberattacks such as man in the middle attack, false data injection, attack on transformer tap changer settings and replay attack in transmission line protection relays have been done based on deep learning trained by measured current and voltage samples. The need for training data sets and the lack of real and adequate attack conditions data is the major problem with this method.

In [39], multi-agent control approach is applied during false data Injection, man in the middle attack, replay attack and integrity attack to increase the security of overcurrent protection relays. The author used multiple controllers and investigates the sequence of accidents along with the integration of physical and cyber data received from synchrophasor. Applying large number of logical operators in algorithm and multiple telecommunication routes is the main drawback of proposed method. In [40], false data injection attack in overcurrent relay settings is detected. Monte Carlo simulation is done to train data sets and load conditional commands using the rough classification method. The proposed method does not implement the principles of adaptive protection based on the online set points changing of the relays. In addition, the penetration of smart

⁹ Expected Load Curtailment

grids and renewable energies has not been taken into account. In [41], protection schemes vulnerability based on telecommunication systems in transmission line distance relay has been investigated during denial of service attack and false data injection. The physical telecommunication methods proposed in this reference are not effective enough to deal with false data injection cyber-attacks. In [42], direct topological cyber-attacks such as line-addition attack, line-removal attack, and line-switching attack has been introduced with the aim of power system disrupting. A new meta-heuristic optimization algorithm such as natural aggregation algorithm is applied to detection. Author shows the impact of a single cyber-attack on power system, and it is better to implement coordinated cyber-attacks to several points.

This paper implements a standard power system during common types of cyber-attacks to present the induced dynamic effects on the electrical signals.

3 | CYBER ATTACK MODELLING

According to the report published by NERC regarding the improper operation of protective relays, more than 20% of the misoperation in this section occur due to malfunctions of circuit breakers and relays [43]. A cyber-attack is one possible reason for an unusual operation of the relays due to penetration of hackers in vulnerabilities of the software and information channels in power system. In such a case, it is very important to investigate the operational behaviour of power system. Among different cyber-attack, injection attacks also known as false data injection attacks (FDIA) include response injection, measurement injection and command injection, DoS attacks and reconnaissance attacks are most possible. Command injections are grouped into three categories as malicious state command injection (MSCI), malicious parameter command injection (MPCI), and malicious function code injection (MFCI). This paper is modelling three types of cyber-attacks as below.

3.1 | Altered actuator state in circuit breaker

It is an MSCI attack that changes the actuator state from on to off or vice versa. This action is done continuously and will also affect the system behaviour and make changes into the physical devices. Usually, the attack on the switching operation or the tripping command sent by the relays, which is change the state of circuit breakers, is known as altered actuator state attacks.

The main objective of the external attacker in a switching sabotage attack is to gain access to the functionality of the circuit breakers. After the occurrence of such attacks, the generators and substations are isolated from the rest of the system and the communication between the areas is disconnect, and therefore the power balance and the constraint related to the amount of power flow between the areas are disturbed. The injected signal of such attack for a specific circuit breaker is a vector such as

$S(x, t) \in \mathbb{R}$ and is formulated as (1).

$$S(x, t) = [y(t) \times Bn] + \text{trip signal of overcurrent relay} \quad (1)$$

where $y(t)$ is a binary pulse signal that generate by intruder and Bn is random Bernoulli binary number. The probability of zero in Bernoulli binary is set near to half. This paper is used MATLAB/Simulink software to generate such a signal that would be launched by a hacker in reality.

3.2 | Altered control set points in overcurrent relay

Here the attacker changes the parameters and it causes the system to perform incorrect control actions. Changing the set points leads to other changes in the system behaviour. This kind of attack is an MPCII one that changes in the set points of the tank in the control schema of the feedback control system or in protection relay setting. This event may trigger other critical response signals and subsequently leads to sudden failure of the system. One major type of attacks on data integrity called altered control set point attack is changing the predefined value set points of control, which causes to digital protection relays malfunction. In these attacks, the intruder hacks the control and protection algorithms of the digital relays by accessing the communication channels and directs them as desired. The final effect of these attacks is the incorrect operation of circuit breakers, the unexplained disconnection of power transmission lines without the occurrence of any physical fault, or keep the power lines in service during the short-circuit.

Therefore, the modelling of the altered set point values attack with the aim of manipulating the integrity can be set as changing the values of the pickup current of the relay. PS^a is the spurious relay pickup current setting that is injected by the attackers, the compromised setting can be written as (2);

$$PS^a = \{ [y(t) \times Bn] + [(1 - Bn) \times (y(t) - y(t - \mathcal{T}_0))] \} \quad (2)$$

where $y(t)$ is a pulse signal with desired amplitude that generate by intruder, Bn is random Bernoulli binary number and $\mathcal{T}_0$ is a delay time. The probability of zero in Bernoulli binary is set near to one. The overcurrent relays, which control the act of isolation of faulty lines from the system without disturbing the healthy lines, are placed at both ends of each line. Thus, the number of directional over-current relays (DOCRs) in an electrical power system is twice the number of the lines. A DOCR has a single input in the form of ac current. While a relay starting up, it sends a message to order a corresponding circuit breaker (CB) cleaning out the fault. The output of the relay is normally an open contact, which changes over to closed state when the relay trips under fault current. The main target of this section is about digital overcurrent relay cyber-attacks.

Per unit current signals at each node is obtained from the current transformer (CT). The current measured by the current transformers is first fed to the protection relays and then

compared with the predetermined threshold values. The result of the comparison is fed to the flip-flop and then a logical AND operator prepares the output signal into double block. This block converts the input to the data type and scaling of the output. At the end stage a delay block simulates a circuit breaker operation delay of 3 to 5 cycles. Usually, there is a very specific method for making decisions by relays, which consists of comparing the values measured by CT with the calculated values of the thresholds that are set and loaded on the relay. In this logic, if the input current is more than the pick-up values, the relay detects the abnormal state and sends a trip signal to open the CB. In this article, the overcurrent relay is used for three phases. IEC 60255 is calculating the nominal operation time of the relay closest to the fault location as (3):

$$t = TMS \times \frac{0.14}{(PSM^{0.02} - 1)} \quad (3)$$

where TMS is time multiplier setting and PSM (plug setting multiplier) is the ratio of fault current to actual pickup current. Usually, the selection of decision threshold or pick-up values is based on a specific principle, and its value is a number between the maximum load current of that feeder and the minimum short-circuit fault current of the system, and is expressed as (4):

$$I_{load-max} \leq I_{pickup} \leq I_{fault-min} \quad (4)$$

It is obvious that pick up current settings for the relays should be above the feeder load currents and not the bus load currents. In fact, one should consider the maximum possible loading conditions, to decide conservatively pick-up current settings. A rule of thumb is to set the pick-up current at 1.25 times maximum load current. Another 'rule of thumb' is to limit pick-up current to 2/3rd. of the minimum fault current. This decides the range available for setting relay pick-up. The selection of the threshold value should respect the practical limit of the relay setting, which is usually between 50% and 200% of the nominal current [44].

3.3 | Denial of service attack in overcurrent relay

Reconnaissance attacks and DoS attacks come under the category of network attacks. DoS attack is planned by hackers to prevent relays from operating at the required time. This type of attack overrides a system's communication resources so that the system becomes unresponsive to other service requests. In such situation the attacker affects the availability of the obtained information by occupied too much of communication channels capacity and sending excessive traffic. DoS attack can be modelled in two forms as follows: In the first one, DoS attack continuously blocks the performance of relays during the fault and emergency situations by transmitting malicious information to the targeted IED. Therefore, the relays do not perform their main mission of sending a timely command to

the circuit breakers in order to reduce the severity of accidents. In this case, the function logic defined in the relay is not sent to the circuit breakers when any fault occurs. In the second case of the DOS attack, the attacker prevents the relays from reacting and sending appropriate commands when faults occur for a limited period of time. In other words, this type of attack is implemented by creating a delay (not sending an instant command) in the trip signal to the circuit breaker when a fault occurs. The injected signal of such attack is expressed as (5):

$$S = e^{-T_0 t} + \text{trip signal of overcurrent relay} \quad (5)$$

where T_0 is a positive scaler that specified by attacker somehow the protection relay output is sent to the circuit breaker with delay. Here it is possible that the delay of the sent signal is so much that it is applied to the circuit breaker after the fault is cleared. This paper is investigated the effect of second type DoS attack described above. All three mentioned cyber-attacks are modelled using MATLAB/Simulink software.

4 | WAVELET TRANSFORM AND PROPOSED METHOD

In the early 1980s, coinciding with the advent of wavelet analysis, its application rapidly developed in various fields such as speech and image signal processing. Wavelet analysis is actually a signal windowing method with adjustable ranges that uses long time windows and short time windows to accurate analyse of low frequency and high frequency signals respectively. The most important advantage of wavelet analysis is its ability to perform local analysis of the studied signal. The wavelet is basically a waveform with limited amplitude and zero average value, which in contrast to the sinusoidal waveform as the basis of Fourier analysis with an infinite range from negative to positive and the predictable waveform has an irregular and asymmetric nature. In Fourier analysis, the original waveform is converted into several sinusoidal waveforms with different frequencies, but the output of the wavelet analysis is a number of shifted and scaled signals from the mother wavelet. The main application of wavelet analysis, due to its nature, is the analysis of sudden and sharp changes. [45]. This analysis decomposes the signal by applying a scale factor (proportional to the inverse of the frequency) on the mother wavelets. Wavelet analysis converts the original signal into approximate and detail signals with different frequency ranges and resolution. This analysis has a high ability to detect fault by extracting detail levels and accessing different frequency bands [46]. The asymmetric discrete wavelet transform (DWT) decomposes every sampled signal $s(s_1, s_2, \dots, s_N)$ into an approximation signal at a certain decomposition level $n(\alpha_n(t))$ and n detail signals ($d_j(t)$) with j varying from 1 to n .

$$\begin{aligned} s(t) &= \sum_i \alpha_n(i) \varphi_{n,i}(t) + \sum_{j=1}^n \sum_i \beta_j(i) \psi_{j,i}(t) \\ &= a_n + d_n + \dots + d_1 \end{aligned} \quad (6)$$

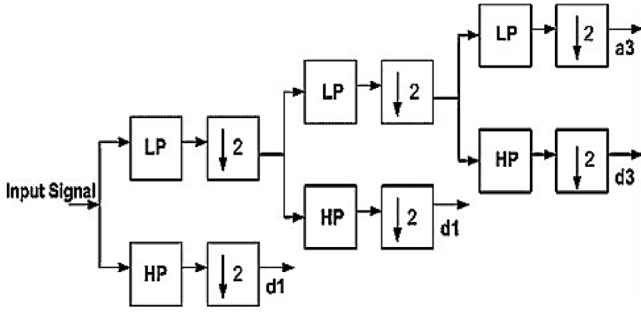


FIGURE 1 Tree algorithm of n -level DWT for a signal

where $\alpha_n(i)$ and $\beta_j(i)$ are the approximation and detail coefficients, respectively, $\varphi_n(t)$ and $\psi_j(t)$ are the scaling function at level n and wavelet function at level j , respectively, and n is the decomposition level. a_n is the approximation signal at level n and d_j is the detail signal at level j [47]. Based on the wavelet theory, the scaling function is equivalent to a low-pass filter and extracts the approximate signals, while the wavelet function shows the detail signals as a high-pass filter. Looking at relation (6), applying asymmetric DWT on a signal is considered as the summation of two products of the scale function with the approximate coefficients of the level n and the wavelet function with the detail coefficients at the j levels.

If the sampling frequency of the signal is considered f_s (sample/s), then the detail signals d_j include components of the main signal whose frequency is placed in the interval $[2^{-(j+1)}f_s, 2^{-j}f_s]$ Hz and approximate signals a_n include the low frequency components of the signal, whose range is $[0, 2^{-(n+1)}f_s]$ Hz.

DWT N -level tree algorithm for a signal is shown in Figure 1. In this article, discrete wavelet analysis is applied to the signal by using the `m-file` commands such as 'wavedec' and 'wrcmf' in the MATLAB environment.

Today, the wavelet is considered a powerful tool in the field of signal processing. According to the nature and shapes of wavelet types, there is a good compatibility between the wavelet family (for example, dbN) and the types of power grid disturbances. It is worth mentioning with increasing N , the cost of implementation increases and the complexity of the design doubles. Therefore, it is needed to select the first wavelet family with ability of detection by trial and error. The next important point is the direct relationship between the volume of calculations and the number of analysis levels, in order to reduce it, the first level with the correct detection capability must be selected.

Considering the growing trend of cyber-attacks in power networks, it is necessary to present a new method and novel index based on detail wavelet signals energy concept. The energy of a discrete signal $X = (x_1, x_2, x_3, \dots, x_N) \in R^N$ is calculated according (7) [48] and cyber-attack detection index (CADI) is defined as (8):

$$E_X = \sum_{i=1}^n |x_i|^2 = x_1^2 + x_2^2 + \dots + x_n^2 \quad (7)$$

$$\text{CADI} = \frac{E_W(x)}{E_O(x)} \quad (8)$$

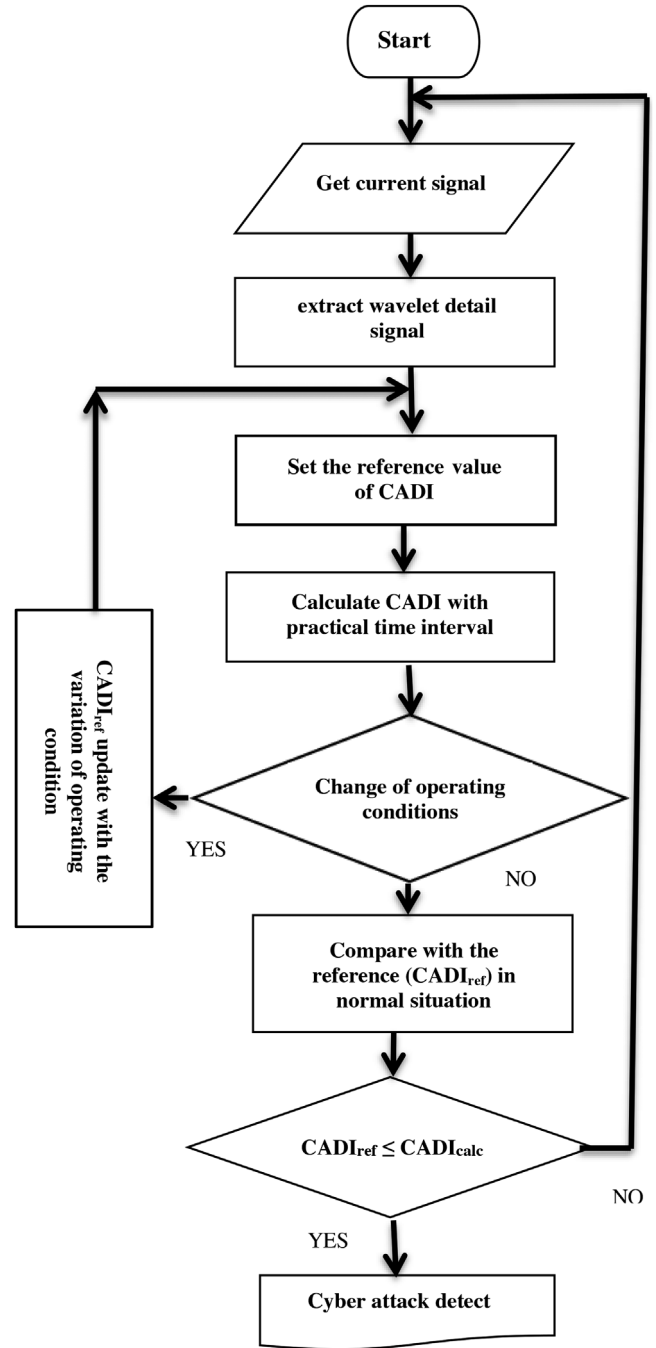


FIGURE 2 Proposed flowchart for cyber-attack detection

where n is the sample number of signal, E_X is the x signal energy, CADI is the cyber-attack detection index, $E_W(x)$ is wavelet signal energy at a certain detail level and $E_O(x)$ is original signal energy. The flowchart of proposed detection method is illustrated in Figure 2.

The CADI_{ref} value should be determine in the system alternatively with the variation of operating condition of unit commitment and updated in the flowchart. The reference value of CADI should be calculated for any cyber protected overcurrent relay and uploaded to the algorithm. So implementation

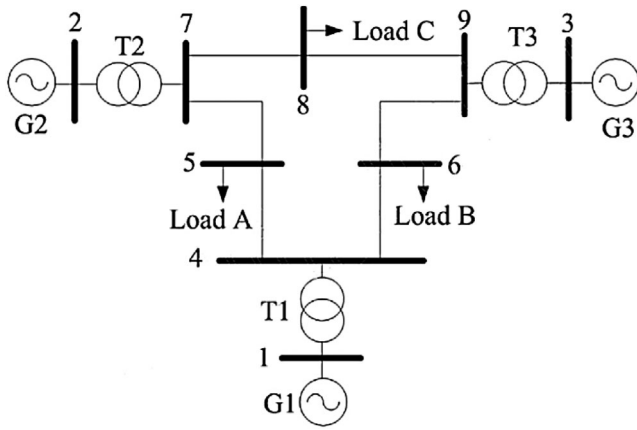


FIGURE 3 WSCC 3-machine 9-bus

of such an adaptive determination plays an important role to distinguish faults and cyber-attacks.

5 | UNDER STUDY TEST SYSTEM

The standard IEEE model of three machines and nine buses is considered for simulating cyber-attacks (Figure 3).

There is a total load of 315 MW in the network, and three generators with numbers one, two, and three with the capacity of 247.5, 192, and 128 MVA and voltage levels of 16.5, 18, and 13.8 kV are responsible for supplying the existing loads. The first generator as an infinite bus is connected to bus number one and generators 2 and 3 are connected to the second and third buses, respectively. Three 100 MVA step-up transformers with a conversion ratio 18 kV/230 kV, 13.8 kV/230 kV and 16.5 kV/230 kV are responsible for matching the output voltage of the generators to the transmission level voltage.

The base voltage is 230 kV. The basic voltage level for generators is considered as the same with the low voltage side of transformer, and therefore, for the transformer located between the two buses #2 and 7, it is 18 kV, the transformer between the two buses #3 and 9 is 13.8 kV, and for the last one between two buses # 1 and 4 is 16.5 kV.

The main task of circuit breakers and relays is to disconnect and connect loads, feeders and transmission lines in different operational conditions. By receiving the current and voltage values sensed by the advanced measuring infrastructure installed throughout the network and substations, the relays send the appropriate command to the circuit breakers. In this article, the hacker targets the digital relay between buses 4 and 5 and imposes various types of cyber-attacks on the system and thereby takes over the circuit breaker function for the middle part of the network on the transmission side, of this path and tries to disrupt it.

The frequencies of the studied system is 60 Hz and in total three loads with powers of 100, 125 and 90 MW are connected to buses number 8, 5 and 6, respectively. Six transmission lines connect the buses in this network for power flow. A summary

of information including load data and transmission line data is available in Tables A1 and A2, respectively in the appendix [49].

6 | SIMULATION RESULTS

The standard IEEE model of three machines and nine buses is considered for simulating in MATLAB/Simulink environment. The sample time and sampling rate of this simulation is 5×10^{-5} s and 20 kHz respectively. A 3-phase fault to ground and three common cyber-attacks such as altered actuator state in circuit breaker attack, altered controlled set points in overcurrent relay attack and DoS attacks are implemented for the assumed network and the dynamic effects of faults and attacks are analysed separately. In this article, it is assumed that the hacker controls the performance of a substation, which is somewhat consistent with the reality where the attacker has limited information about the system. Cyber-attacks manipulate the relays in vulnerable substations. Because of the relays substation failure, the affected power system experiences an unexpected dynamic behaviour which may result in system blackouts. Different scenarios along with their dynamic effects on the system are examined below.

6.1 | Impact of a three phase symmetrical fault

First, the effect of three-phase to ground short circuit fault is analysed as its worst type. As it is known, overcurrent relays by continuously receiving changes in current through CTs and performing internal calculations detect the increase in current more than usual and clear the fault by sending a trip command to the circuit breaker. Generally in situation of transient current faults and in the absence of an automatic recloser, the operator must be present at the switchboard and reset the current lock-out relay to allow reconnection of the breaker. However, there is no such requirement for voltage faults and the system is ready to reconnect as soon as the fault is resolved. In permanent current faults, it is possible to reconnect the switch after the failure source is eliminated. This section simulates a permanent current fault. Therefore a three-phase fault to the ground with $R_g = 0.01$ ohm (ground resistance) and $R_{on} = 15$ ohm (fault resistance) is applied at $t = 2$ s on bus number 7 and permanently remain until the end of simulation at $t = 10$ s. The relay related to substation No. 7 sends the disconnection command to the circuit breaker located between buses No. 5 and 7 upon detecting the increase in the current level over its defined value. The dynamic treatment of the fault is shown in Figure 4. The fault current in bus 7 increases by 10 p.u. (Figure 4a) and therefore the overcurrent relay quickly detects this unusual situation. The disconnection process is completed after 4 cycles, which is within the practical standards of the power system.

The load angles of the three generators with their fluctuation due to the rapid rise in the current are shown in Figure 4b. Generator no. 1 acts as infinite bus and its load angle is close to the zero.

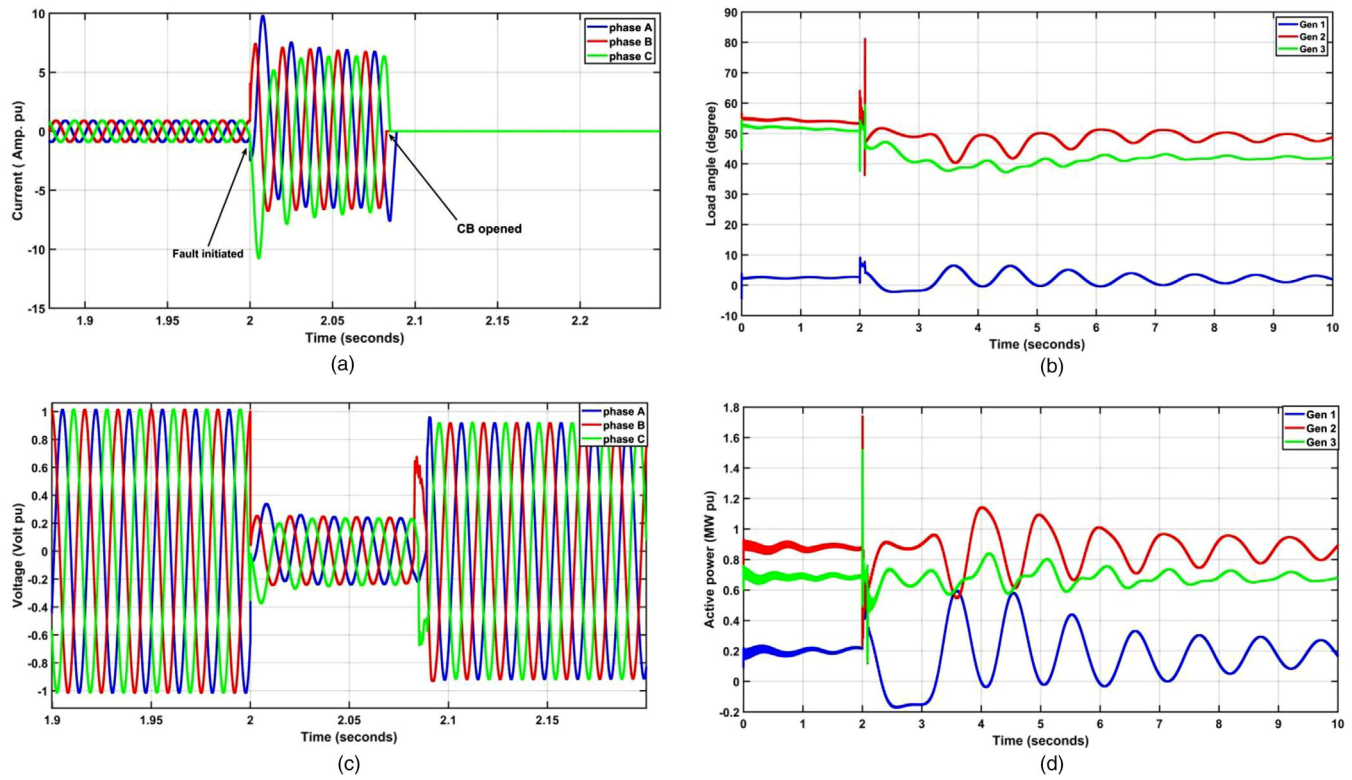


FIGURE 4 (a) Line currents from bus 7 to 5. (b) Load angle variation of three generators. (c) Measured voltages at substation 7. (d) Generators contribution of active power injection

The three-phase voltage drops toward zero in the fault state and converges to the previous value after it is resolved.

The delivered active powers of generators also have fluctuation as the results of this fault (Figure 4d). Such fluctuations in load angles and active powers are gradually damped and power system returns to stability after clearance the fault.

6.2 | Impact of an altered actuator state attack in circuit breaker

Today, power systems can be controlled remotely by the operator using modern communication and telecommunication technologies. Therefore, when an attacker gains access to the central control system of the substations or infiltrates the communication channels, can direct the on/off command output from the relay to the circuit breaker and change it as desired. In this case, by disrupting the process and manipulating the channels, the hacker starts the altered actuator state attack. In this case, the hacker sends the pre-designed commands to the communication channels leading to the circuit breaker, and they are the remaining latch there. As soon as the status of the circuit breakers changes, these commands are applied to the system. For example, if the breaker is opened due to a fault, the hacker command could be successive closing and openings. It should be noted that this type of cyber-attack is not only related to the occurrence of the fault and is possible in any case of system

manoeuvring. In the following, an altered actuator state attack is applied in the form of remote control at substation number 7, and then the dynamic changes of the signals are monitored and checked. In this attack, after controlling the overcurrent relay installed in the substation, the attacker send some random false ON/OFF commands after $t = 4$ s to the circuit breaker located between bus 5 and 7. In this case a permanent fault mode from $t = 2$ s to $t = 10$ s can be examined. Line currents due to the frequent switching and continuous connection and disconnection of the circuit breaker between two buses 5 and 7, severely fluctuate during this period. The inrush currents registered in the relay is rises up to 10 p.u. in such situation. The quality of changes in line currents during the attack is shown in Figure 5a. The difference in the level of current rise in diagram at different times is due to the synchronization of the switching moment and the phenomenon of zero crossing.

It could be clearly seen in Figure 5b the load angle of the generators has fluctuated greatly in permanent fault because of repeating the operation of switching of the circuit breaker. This issue leads to the hunting effect in motors and generators. As it is obvious from the results, the system becomes unstable in permanent fault. Three-phase voltages in Figure 5c show the variation of signal in such situation.

Figure 5d shows the output power of the generators. Due to the opening of the circuit breaker between bus 5 and 7, there is no path to feed the 125 MW load through generator number 2, and in this case, only infinite bus and generator number

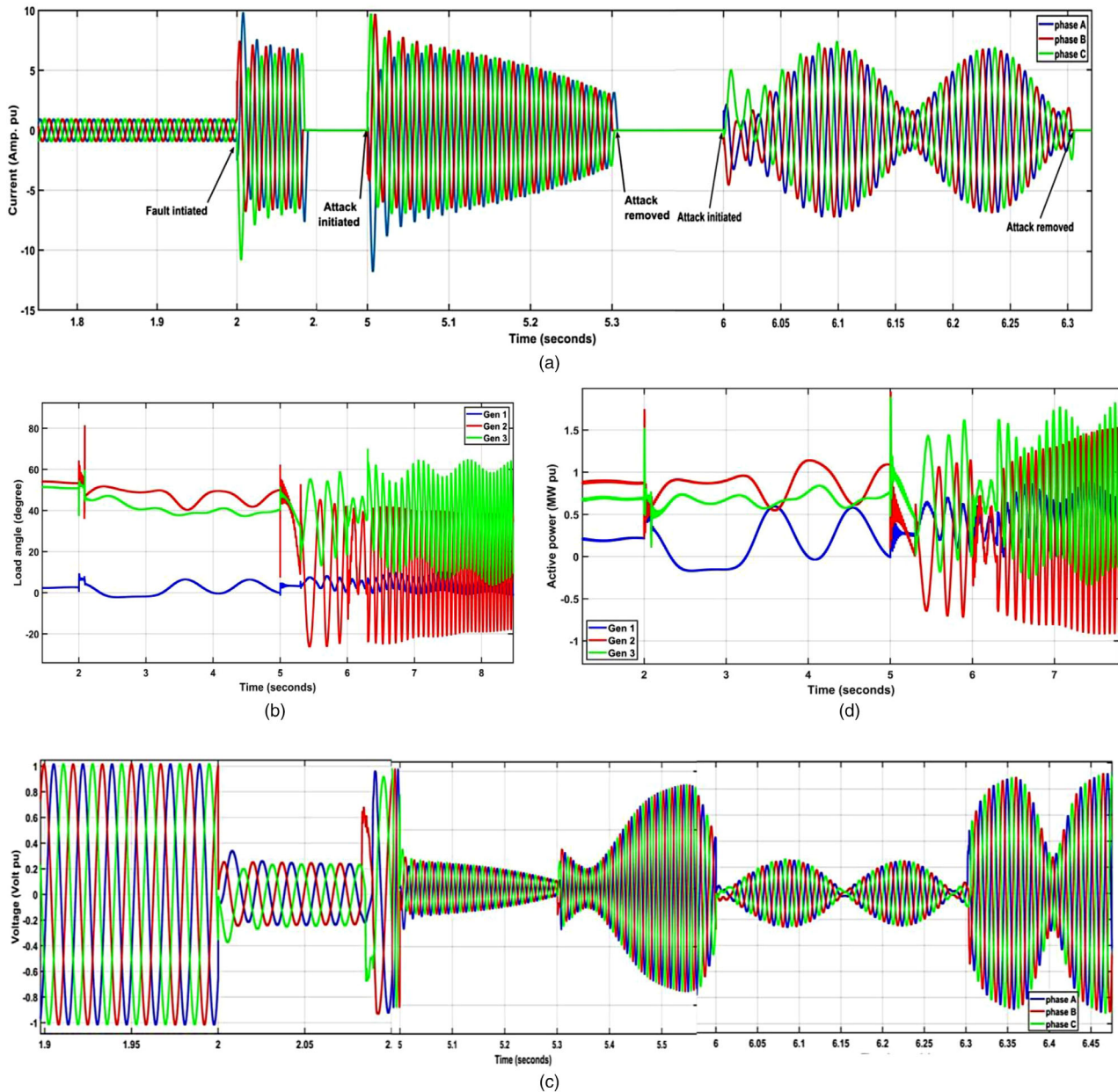


FIGURE 5 (a) Line currents from bus 7 to 5. (b) Load angle variation of three generators. (c) Measured voltages at substation 7. (d) Generator contribution of active power injection

3 supply this load. Therefore, with the reduction of the delivered power by generator 2, generator number 3 is required to generate more power to supply loads and balanced the grid. In this type of attack, which is known as altered actuator state, the circuit breaker is opened and closed several times for no reason and intentionally by the hacker. So it would be seem that this attack has the same function as a recloser. It should be noted here that this protection device operates at a limited repetition and predetermined duration. In return cyber-attack does not follow such a pattern. In general, the performance of the recloser is associated with maintaining the stability of the system, but in a cyber-attack, the goal is to destabilize the system.

6.3 | Impact of an altered control set points attack in overcurrent relay

During power system commissioning, a settings list is used to configure the relays, in which the threshold values are set. In order to detect faults in the network, the settings of the relays must be done accurately so that it can send the necessary command to the circuit breaker when needed. Any manipulation in the threshold values of the relays is an altered control set points attack. If the hacker changes the threshold values upwards, it is most likely that the short circuit fault is not detected by the relay and the circuit breaker does not function in such abnormal conditions and does not isolate the faulty part.

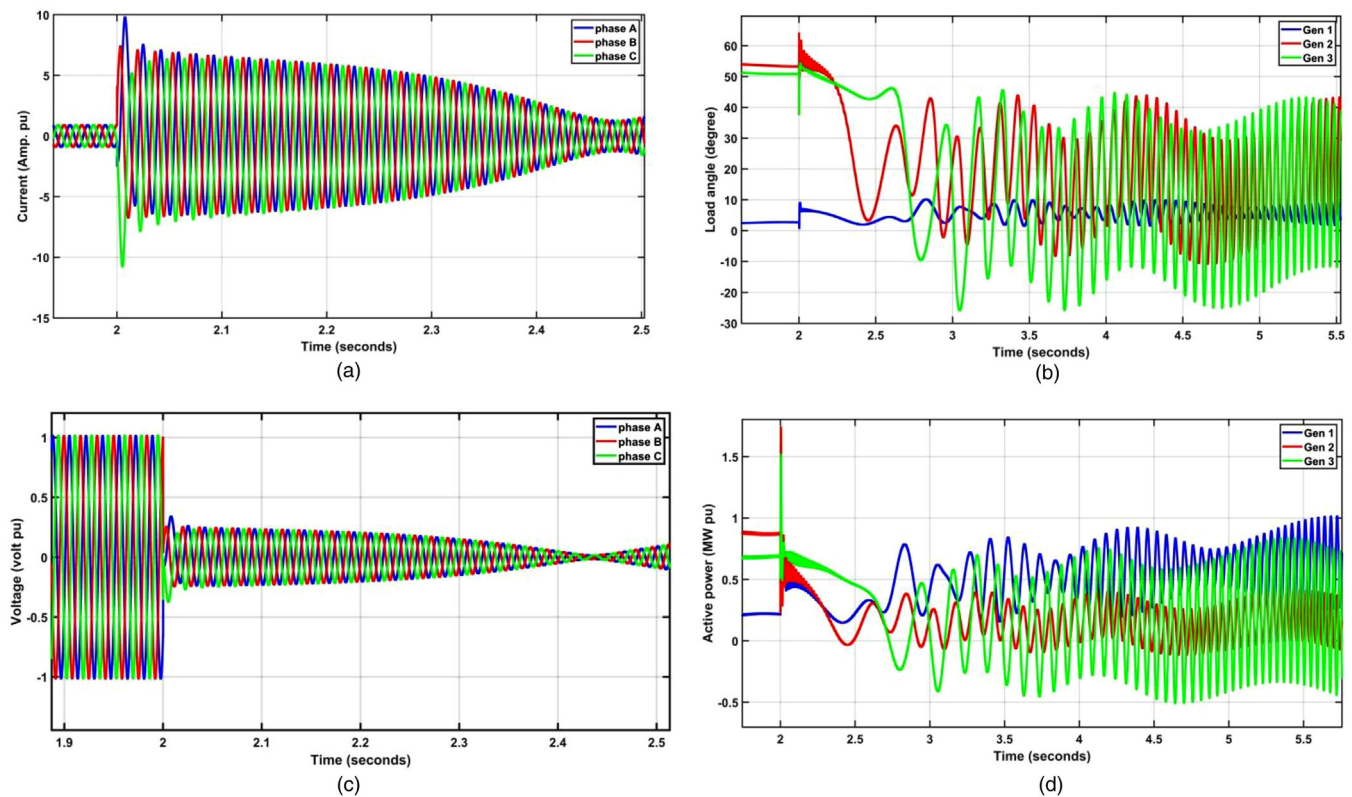


FIGURE 6 (a) Line currents from bus 7 to 5. (b) Load angle variation of three generators. (c) Measured voltages at substation 7. (d) Generator contribution of active power injection

On the other hand, if the threshold values are reduced due to attack, it is possible that any increase in the system load will be considered as a fault and the trip command will be sent by relay. In this section such above mentioned attack is implemented for the standard IEEE model of three machines and nine buses system. According to the standards, the threshold value setting is twice (equal to 200%) the nominal current that flow through the line 5–7 for proper detection. Therefore, in this type of attacks, it is assumed that the hacker increases threshold current value from 2 to 15 p.u. by invading and penetrating the settings of the relays and changing their predefined. Then a three-phase fault to ground on bus #7 is applied at $t = 2$ s and terminated at $t = 10$ s. The high inrush current induced by the fault is not detected through the overcurrent relay, and its main cause is a sabotage attack and changing the relay settings (Figure 6a).

As a result, the circuit breaker continues to operate and does not trip. Therefore, the fault remains continuously in the system and the power angles of the generators fluctuate greatly, which can lead to the overall instability of the system (Figure 6b).

The voltages at bus 7 remain near zero during the fault (Figure 6c) and then oscillate around it during the fault occurrence.

The general increase in the power of generators can cause them to violate the permitted power limit and put them in a

state of severe overload (Figure 6d). The generator 3 delivered power is increasing divergently.

6.4 | Impact of a denial of service attacks in overcurrent relay

A DoS attacks usually start by creating a delay or a complete prevention in sending commands from digital relays to circuit breakers. During the execution of these attacks, an intruder can be set the performance of the relays in three modes of blocking, delaying, or rejecting, and the result of a successful attack can lead to irreparable damages in the operation and dynamic treatment of the system.

To execute a DOS attack, a three-phase fault is applied to ground on Bus 7 at $t = 2$ s. Due to the presence of protective algorithms, the overcurrent relay detects the occurrence of the fault immediately, but the activation of the denial of service attack causes that the disconnection command to be sent with a delay of 0.4 s and the circuit breaker operates after 0.48 s. Basically, the maximum acceptable delay when issuing the command to trip the circuit breaker in the power network, which is calculated as the sum of computing and communication delays, is considered to be 4 cycles. In this case of simulation, because the applied delay is more than the standard amount, it causes strong fluctuations in the dynamic behaviour of the system. It is

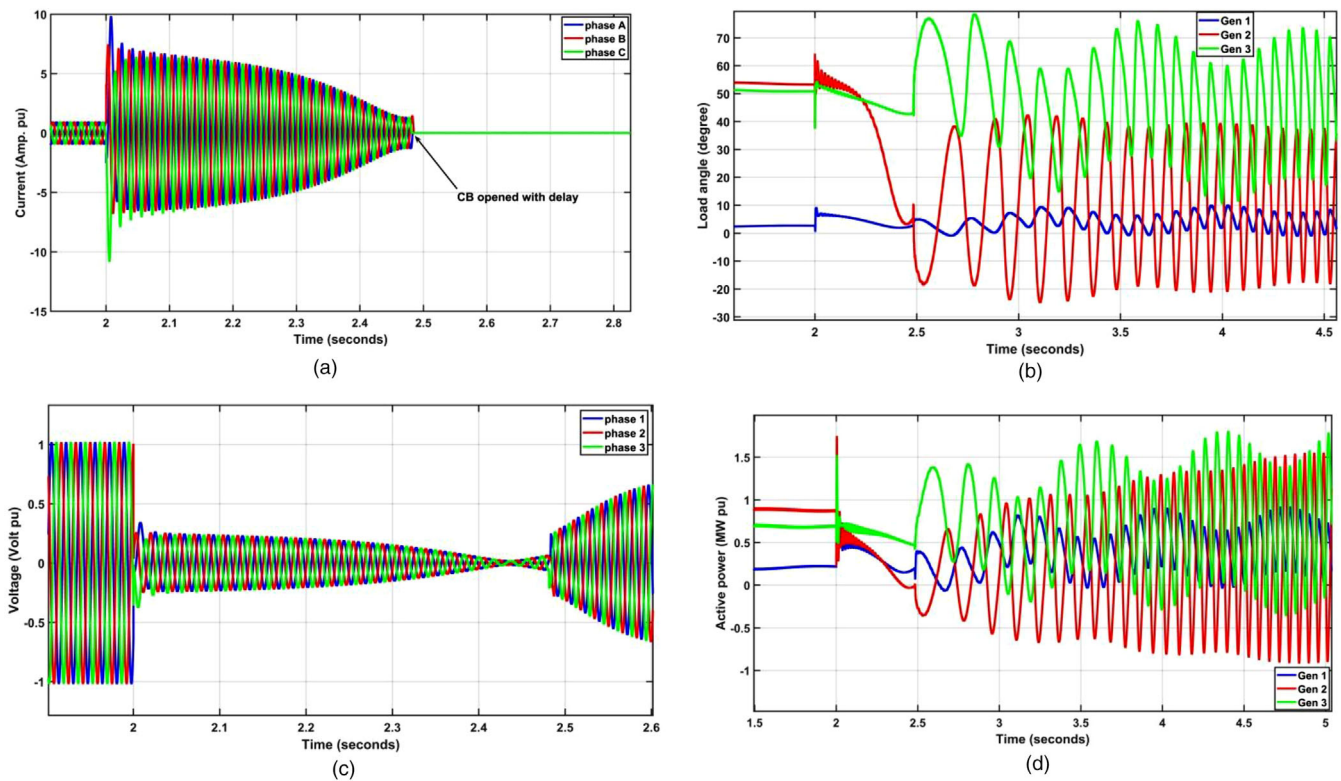


FIGURE 7 (a) Line currents from bus 7 to 5. (b) Load angle variation of three generators. (c) Measured voltages at substation 7. (d) Generators contribution of active power injection

clear that if the fault persists for more than the standard defined time (0.2 s) and the protection equipment does not work on time, the power grid will experience unstable conditions compared to the mode of clearing the fault in time [50]. The line current from bus 7 to 5 is shown in Figure 7a.

By comparing the output signals of the system in the two cases of DOS attack and three-phase to ground fault (in Section 6.1), the existence of strong fluctuations with a higher amplitude in the load angle of the generators (Figure 7b) and the delivered power (Figure 7d) during the execution of such attacks is clearly evident and consequently system becomes unstable.

As it is clear in the Figure 7c, when the line voltage is zero for a long time due to the delay injected by the attack in sending the trip command and the lack of operation of the circuit breaker, catastrophic damages is imposed to the electrical equipment for the sake of basic insulation level (BIL) violation.

It should be noted here that the effective cyber-attack is occurred when a hacker identified the power system structure and disable other backup protections. The important point in all these scenarios is to detect the failure state of the relays with a cyber-attack that can be recognize by online monitoring of network dynamic parameters in a control centre before causing damages or blackouts. For this purpose, some relay manufacturers use additional protection circuits to detect relay malfunction

To validate the result a comparisons has been done between some pervious literatures in this aspect. As shown in ref.

[36], the occurrence of cyberattacks causes system instability and extreme oscillation in the various waveforms of signals, confirming the simulation results obtained in this paper.

6.5 | Detection of cyber-attack by proposed method

To detect various cyber-attacks, wavelet analysis of different signal has been considered. Among the available signals, the line current was selected as the best option for detecting and classifying cyber-attacks. The application of electrical signals, which are usually used for control purposes and conditions monitoring in the system, is completely justified from a practical, technical and economic point of view, and therefore the costs related to the design and technical issues caused by the installation of new equipment and sensors are completely eliminated. Comparison the results show that the direct implementation of n-level DWT of the line current in normal, faulty and cyber-attack situation does not have the ability to provide comprehensive and discriminating information for accurate diagnosis and only shows slight changes in the sampled signals. According to the literature review, Daubechies family has been the most of applications in power systems analysis. Further, often db4 have been the satisfactory results than the other mother wavelets of Daubechies family [51]. The best decomposition level of wavelet analysis is selected in a trial and error manner with the aim of clear and

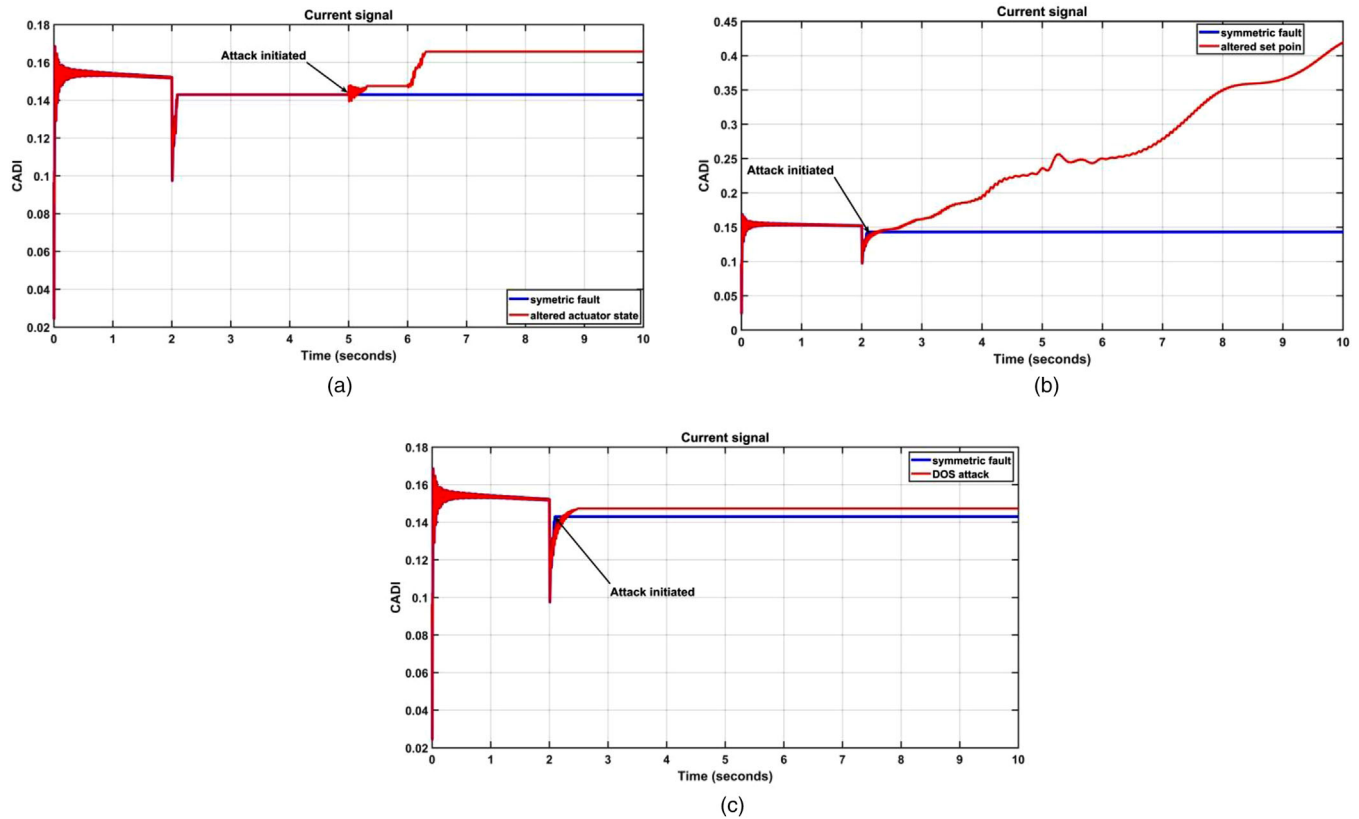


FIGURE 8 (a) CADI curve in altered actuator state attack in compare with symmetric fault. (b) CADI curve in altered set point attack in compare with symmetric fault. (c) CADI curve in DOS attack in compare with symmetric fault

comprehensive discrimination ability as well as having the minimum volume of computation. Figure 8a–c shows CADI curves based on current signal for considered different cyber-attacks in comparison with symmetrical fault. The db4 wavelet family at ten levels has been executed. CADI at the seventh level was selected as the best option for detection and classification. As it is obvious from Figure 8, the level of CADI curve in cyber-attacks is always higher than symmetrical fault. This problem can be considered as the basic principle and basis of the proposed diagnosis method.

To validate the use of wavelet analysis in detecting cyber-attacks, comparison between the results of employing wavelet signal entropy in different modes in ref. [52] can confirm the use of wavelet energy signal concept in the proposed method of attacks detecting.

6.6 | Sensitivity analysis of proposed method

The comprehensive coverage of different conditions and the ability to implement in industrial environments are considered as the main capabilities of the described method. To evaluate sensibility of suggested method, various signal such as load angle of generators, bus voltage and delivered active power signal could be implemented at different sample rate. In such situation, the nature of CADI curve has the same results as before.

So it can be concluded that the proposed novel method is robust to changing the sampling signal and its rate. Figure 9a–c shows CADI curves based on load angle signal for considered different cyber-attacks in comparison with symmetrical fault. Results show the current signal detects cyberattack more quickly and clearly than other signals and the inertia of the proposed algorithm is less in this case.

In a slight case of the fault or cyber-attack while current could be reach to ten times of the current transformer (CT) nominal current such as simulation in this paper, the CT performance has no impact on suggested detection method because the amount of current increasing is not harmful and does not limit in relay input. But low-ratio CTs transmit proportionally less signal energy to the relay when the primary current increases impressively. In an extreme case of the fault or cyber-attack while current reached to 1000 times of the CT rating; only a small percent of this current near the value of slight case, in the form of short spikes would be delivered to the relay. However, saturation of relay input CTs may be neglected for the following reasons [53]:

- The secondary current is dramatically reduced under severe saturation of CTs.
- CT Saturation makes the secondary current symmetrical and eliminating the risk of exposing the relay input CT by collapsing dc components.

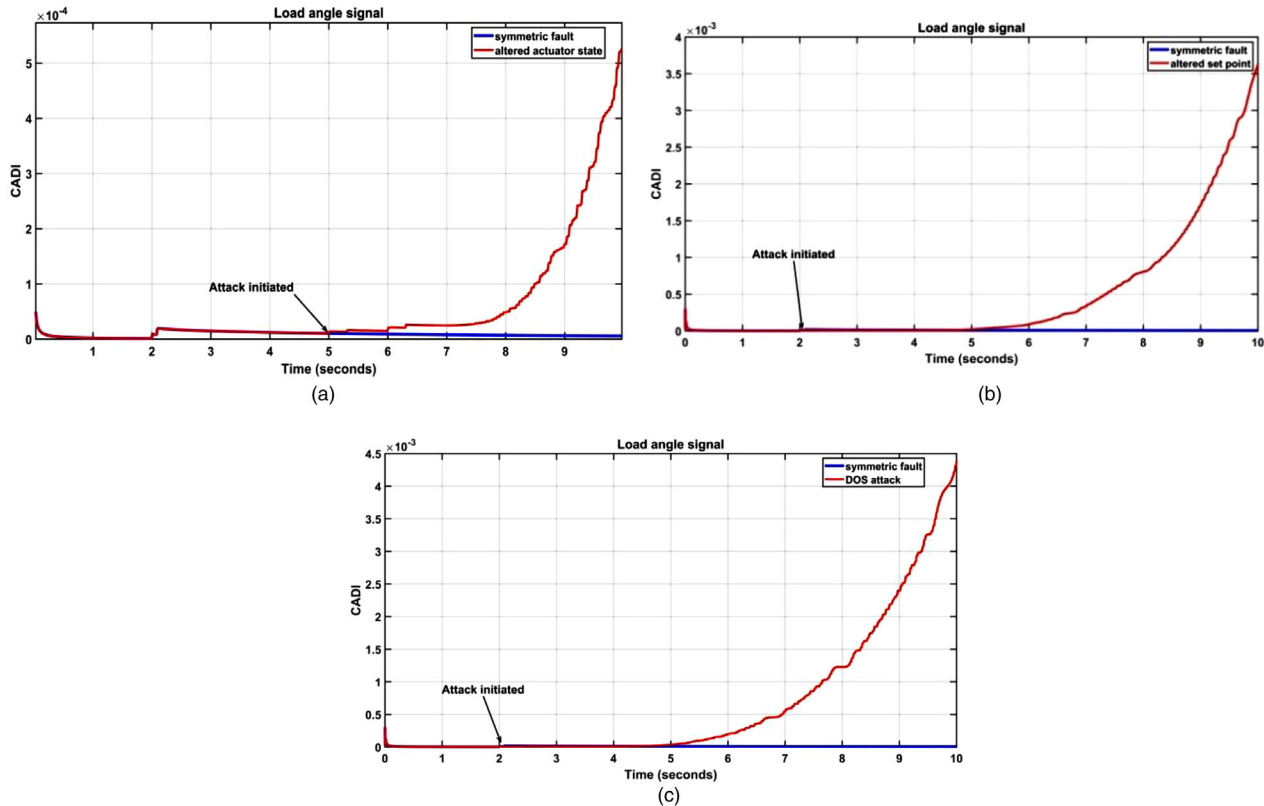


FIGURE 9 (a) CADI curve in altered actuator state attack in compare with symmetric fault. (b) CADI curve in altered set point attack in compare with symmetric fault. (c) CADI curve in DOS attack in compare with symmetric fault

- The secondary current is appearing as short lasting spikes. This limits the flux in the cores of the relay inputs CTs.

Therefore, it can be said that CT saturation and its performance in both modes of cyberattacks with low and very high current changes have no effect on index calculation and attack detection.

Investigation of CADI curves show that the proposed detection method is robust to variation of sampling frequency, sampling signal and CT saturation. In these surveys, CADI curve is always above the faulty one, so it is proved that the proposed method is robust.

The results of the proposed detection method are examined in five different operating points. Each of the three types of cyberattacks is applied ten times to the system with various scenarios. Considering cyberattacks and three-phase short circuit fault to ground, a total of 155 simulations are performed. The result for relay and the performance of cyber-attack detection algorithm is shown in Tables 1 and 2, respectively.

7 | DISCUSSION

The development of cyberattack detection systems for improving the cybersecurity of protective relays is at its primary stage. Despite the promising results obtained in this article, several

TABLE 1 Results for Relay#7

		Predicted	
		Yes	No
Actual	Yes	True positive (TP) = 150	False negative (FN) = 0
	No	False positive (FP) = 0	True negative (TN) = 5

open challenges should be addressed before it can be applied to real systems. The first is related to the scarcity of fault data in power system that is required to predetermine in the cyber-attack detection algorithm. One can overcome this challenge by developing dynamical models that represent the real power grids in time domain simulators and validating the dynamical models with data from the fault recorders. We emphasize that it is impossible to develop cyberattack detection systems for protective relays without access to high fidelity data. Thus, the development of accurate time-domain dynamical models is the essential first step for the advancement of cyberattack detection systems. There are numerous scenarios and practical considerations that should be taken into account before implementing the proposed model in practice. For instance, investigation the impact of scenarios such as lightning strike and other transients on the CADI curve, the existence of short lines, load shedding, sudden outage, load increasing, different fault impedances and

TABLE 2 Performance of cyber-attack detection algorithm

Fault scenario	Type of cyber attack	Accuracy $= \frac{\#TP + \#TN}{\text{total}}$	Error rate (ERR) $= \frac{\#FP + \#FN}{\text{total}}$	Precision $= \frac{\#TP}{\#TP + \#FP}$	Sensitivity (recall) $= \frac{\#TP}{\#TP + \#FN}$
A-B-C-G	Altered actuator state	100%	0%	100%	100%
	Altered set point	100%	0%	100%	100%
	DOS	100%	0%	100%	100%

varying penetration of distributed energy resources should be analysed on the performance of the proposed model. Moreover, several considerations such as different transmission network topologies, communication packet loss, noise in wavelet and the other cyber-attack such as deception and covert attack could be implemented. The feasibility of mentioned cyber-attacks in the actual communication system is the goal for future work. Due to the fact that the proposed method calculates energy based on the data received from the measurement sensors such as CT, so it loses its detection capability if the signal magnitudes at the wavelet algorithm input are changed by intruder.

8 | CONCLUSION

In this article, a novel simplified and randomize modelling has been formulated and performed for three types of common cyber-attack. The major advantage of such modelling is the ability of provide the main concept of cyber threats and its effect on system protection. Detection and distinction of cyber-attack at the beginning step is so hard because of its similar treatment like as symmetrical fault and relay malfunctions on system operation. In recent years, many researchers have been inclined to investigate different angles in this field. This article derived that the wavelet analysis of the current signal has been reasonably effective for diagnosis of cyber-attack which commonly compromise the power system. The proposed detection algorithm is expressed as a combination of discrete wavelet analysis and the concept of latent energy in an analogue signal. At the same time, a new simple index is expressed in this method, which is the ratio of the detail wavelet signal energy at a certain stage to the energy of the main signal. Proposed index has been examined for various signal and cyber-attacks. Results show that the level of CADI curve in cyber-attacks is always higher than symmetrical fault and current signal detects cyberattack more quick and clear than other signals with accuracy of 100% without any false trigger. This conclusion can be used as the basis of cyber-attacks detection. It is worth noting that the proposed unique algorithm is resistant to any changes in the sampling signal, its rate and CT saturation. This method is superior to other detection techniques because is not dependent on operating point, relay protection scheme and cyber-attack type. Due to the fact that in a certain operating point of the system, cyber-attacks have a more severe effect than the abnormal actions of breakers due to unexpected cyber contingencies caused by natural disasters or software malfunctions, so the proposed method could

distinguish them from each other. The algorithm implementation is costly because of the continuous real time condition monitoring requirement.

AUTHOR CONTRIBUTIONS

All authors contributed to conceptualization, data curation, methodology, analysis of the results and discussion, validation, writing-original draft, writing-review and editing of the final manuscript.

CONFLICT OF INTEREST

The author declare no conflict of interest.

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DATA AVAILABILITY STATEMENT

The power grid data that support the findings of this study are openly available in ref. [49].

ORCID

Mohammad Yousefi kia  <https://orcid.org/0000-0003-2204-2781>

Mohsen Saniei  <https://orcid.org/0000-0002-0215-763X>

Seyyed Ghodrattollah Seifossadat  <https://orcid.org/0000-0001-9000-1522>

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APPENDIX A

TABLE A1 Load data

Bus no.	P (MW)	Q (MVAR)
5	125	50
6	90	30
8	100	35

TABLE A2 Transmission line data

Branch	L (H/km)	C (F/km)	R (Ω /km)	L (km)
4-5	1.192×10^{-3}	8.82×10^{-9}	0.0529	100
4-6	1.29×10^{-3}	7.922×10^{-9}	0.08993	100
6-9	2.38×10^{-3}	17.95×10^{-9}	0.20631	100
5-7	2.259×10^{-3}	15.34×10^{-9}	0.16928	100
7-8	1.01×10^{-3}	7.471×10^{-9}	0.044965	100
8-9	1.414×10^{-3}	10.47×10^{-9}	0.062951	100